

CO 250 Spring 2026: Assignment 4 solutions

- A4-1. (a) We first turn this into a maximization problem by multiplying the objective function by -1 to get

$$\max (-3, -1, -4, -1, -5)(x_1, x_2, x_3, x_4, x_5)^T.$$

Since the second and third constraints are inequalities, we need to add/subtract non-negative slack variables to make them equalities. Add new variables x_6, x_7 and get

$$x_2 + x_4 - x_6 = 2$$

$$-x_1 - 2x_2 - 3x_3 - 4x_4 - 5x_5 + x_7 = 6$$

We see that x_3 is free, so we define two new non-negative variables x_3^+, x_3^- and replace x_3 by $x_3^+ - x_3^-$ everywhere in the LP.

Also, $x_4 \leq 0$, but we need non-negative variables. So we define a new non-negative variable x_4^- and replace x_4 by $-x_4^-$ everywhere in the LP.

Also, This yields the following equivalent LP in SEF:

$$\begin{aligned} \max \quad & (-3, -1, -4, 4, 1, -5, 0, 0)(x_1, x_2, x_3^+, x_3^-, x_4^-, x_5, x_6, x_7)^T \\ \text{s.t.} \quad & \begin{pmatrix} 1 & 2 & 3 & -3 & -4 & 5 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 0 & -1 & 0 \\ -1 & -2 & -3 & 3 & 4 & -5 & 0 & 1 \\ 1 & 0 & 1 & -1 & 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3^+ \\ x_3^- \\ x_4^- \\ x_5 \\ x_6 \\ x_7 \end{pmatrix} = \begin{pmatrix} 9 \\ 2 \\ 6 \\ 5 \end{pmatrix} \\ & x_1, x_2, x_3^+, x_3^-, x_4^-, x_5, x_6, x_7 \geq 0 \end{aligned}$$

- (b) We see that x contain free variables. So for each x_i , we need to replace it with $x_i^+ - x_i^-$. Define $x^+ = (x_1^+, \dots, x_n^+)^T$, $x^- = (x_1^-, \dots, x_n^-)^T$. Then we will replace all instances of x in (P) with $x^+ - x^-$ and set $x^+, x^- \geq 0$.

The first set of constraints are inequalities, so we need to subtract a non-negative slack variable for each such constraint. Define s to be the vector of slack variables (dimension of s is equal to the number of rows of A). Then we will add $-Is$ to form equality.

So in standard equality form, the LP is

$$\begin{aligned} \max \quad & c^T x^+ - c^T x^- - d^T y \\ \text{s.t.} \quad & Ax^+ - Ax^- - Is = b \\ & Bx^+ - Bx^- + Cy = f \\ & x^+, x^-, y, s \geq 0 \end{aligned}$$

In matrix form, this becomes

$$\begin{aligned} \max \quad & (c^T \quad -c^T \quad -d^T \quad 0^T) \begin{pmatrix} x^+ \\ x^- \\ y \\ s \end{pmatrix} \\ \text{s.t.} \quad & \begin{pmatrix} A & -A & O & -I \\ B & -B & C & O \end{pmatrix} \begin{pmatrix} x^+ \\ x^- \\ y \\ s \end{pmatrix} = \begin{pmatrix} b \\ f \end{pmatrix} \\ & x^+, x^-, y, s \geq 0 \end{aligned}$$

(c) We prove the contrapositive of the statement: (P) is feasible if and only if (Q) is feasible.

($\Rightarrow$) Suppose (P) is feasible. Then there exists a feasible solution (x, y) . We now construct a solution $(\bar{x}^+, \bar{x}^-, \bar{y}, \bar{s})$ as follows: Let $\bar{x}_i^+ = |x_i|$ for each i , and $\bar{x}_i^- = |x_i| - x_i$ for each i . Notice that this means $\bar{x}_i^+ - \bar{x}_i^- = x_i$. Let $\bar{y} = y$, and $\bar{s} = Ax - b$.

We now check that this is a feasible solution for (Q). For the first constraint, we see that

$$A\bar{x}^+ - A\bar{x}^- - I\bar{s} = A(\bar{x}^+ - \bar{x}^-) - (Ax - b) = Ax - Ax + b = b.$$

For the second constraint, we see that

$$B\bar{x}^+ - B\bar{x}^- + C\bar{y} = B(\bar{x}^+ - \bar{x}^-) + Cy = Bx + Cy = f$$

where the last equality is due to the fact that (x, y) is feasible for (P).

For non-negativity, we see that $|x_i| \geq 0$, and $|x_i| - x_i \geq 0$ since $|x_i| \geq x_i$. Also $y \geq 0$ since y is feasible for (P), and $Ax - b \geq 0$ since x satisfies the constraint $Ax \geq b$.

Therefore, (Q) is feasible.

($\Leftarrow$) Suppose (Q) is feasible. Then there exists a feasible solution (x^+, x^-, y, s) . We claim that $(\bar{x}, \bar{y}) = (x^+ - x^-, y)$ is feasible for (P). For the first constraint,

$$A\bar{x} = Ax^+ - Ax^- = b + Is \geq b$$

where the last equality is due to (x^+, x^-, y, s) being feasible for (Q), and the last inequality is due to $s \geq 0$.

For the second constraint,

$$B\bar{x} + C\bar{y} = Bx^+ - Bx^- + Cy = f$$

since (x^+, x^-, y, s) is feasible for (Q).

And it is easy to see that $\bar{y} = y \geq 0$. Hence (P) is feasible.

A4-2. (a) This is a basic feasible solution for the basis $B = \{3, 5, 7\}$. We see that the vector is non-negative, and

$$\begin{pmatrix} -4 & 8 & 0 & 5 & 3 & 3 & 1 & 3 & 2 \\ 1 & -5 & 4 & 4 & 3 & 1 & 0 & 4 & 3 \\ 2 & -4 & -1 & 2 & 2 & 0 & 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 2 \\ 0 \\ 3 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 9 \\ 10 \\ 6 \end{pmatrix}$$

So the vector is feasible.

The non-basic variables are 0's. The columns $\{3, 5, 7\}$ are linearly independent, as their RREF is the identity.

$$A_B = \begin{pmatrix} 0 & 3 & 1 \\ 4 & 3 & 0 \\ -1 & 2 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

To modify the constraints into canonical form, we need to find A_B^{-1} .

$$A_B^{-1} = \begin{pmatrix} 0 & 3 & 1 \\ 4 & 3 & 0 \\ -1 & 2 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} -3 & 1 & 3 \\ 4 & -1 & -4 \\ -11 & 3 & 12 \end{pmatrix}$$

We now multiply A_B^{-1} on both sides of the constraints to get

$$\begin{pmatrix} 19 & -41 & 1 & -5 & 0 & -8 & 0 & -5 & 3 \\ -25 & 53 & 0 & 8 & 1 & 11 & 0 & 8 & -3 \\ 71 & -151 & 0 & -19 & 0 & -30 & 1 & -21 & 11 \end{pmatrix} x = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

To get the objective function, we calculate y first.

$$y = A_B^{-T} c_B = \begin{pmatrix} -3 & 4 & -11 \\ 1 & -1 & 3 \\ 3 & -4 & 12 \end{pmatrix} \begin{pmatrix} 4 \\ -5 \\ 2 \end{pmatrix} = \begin{pmatrix} -54 \\ 15 \\ 56 \end{pmatrix},$$

Then the objective function is

$$\begin{aligned} z &= [c^T - y^T A]x + y^T b \\ &= \left[(-3 \ 1 \ 4 \ 1 \ -5 \ 9 \ 2 \ -6 \ 5) - (-54 \ 15 \ 56) \begin{pmatrix} -4 & 8 & 0 & 5 & 3 & 3 & 1 & 3 & 2 \\ 1 & -5 & 4 & 4 & 3 & 1 & 0 & 4 & 3 \\ 2 & -4 & -1 & 2 & 2 & 0 & 1 & 0 & 2 \end{pmatrix} \right] x \\ &\quad + (-54 \ 15 \ 56) \begin{pmatrix} 9 \\ 10 \\ 6 \end{pmatrix} \\ &= (-346 \ 732 \ 0 \ 99 \ 0 \ 156 \ 0 \ 96 \ -44) x \end{aligned}$$

- (b) This is not a basic feasible solution. A basis consists of 3 variables, so up to 3 entries of a basic feasible solution can be non-zero. This vector has 4 non-zero entries.
- (c) This is not a basic feasible solution. Suppose it is, then the only choice of basis is $\{1, 6, 9\}$ as they are the non-zero entries. However, the columns corresponding to the basis are linearly dependent, since

$$\begin{pmatrix} -4 \\ 1 \\ 2 \end{pmatrix} + 2 \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix}.$$

- (d) This is not a basic feasible solution, since $x_2 = -1 < 0$.
- (e) This is a basic feasible solution for the basis $B = \{1, 4, 9\}$. We see that the vector is non-negative, and

$$\begin{pmatrix} -4 & 8 & 0 & 5 & 3 & 3 & 1 & 3 & 2 \\ 1 & -5 & 4 & 4 & 3 & 1 & 0 & 4 & 3 \\ 2 & -4 & -1 & 2 & 2 & 0 & 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 9 \\ 10 \\ 6 \end{pmatrix}$$

So the vector is feasible.

The non-basic variables are 0's. The columns $\{2, 4, 9\}$ are linearly independent, as their RREF is the identity.

$$A_B = \begin{pmatrix} 8 & 5 & 2 \\ -5 & 4 & 3 \\ -4 & 2 & 2 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

(Note: Alternatively, it is also a basic feasible solution for the bases $\{3, 4, 9\}$, $\{4, 5, 9\}$, $\{4, 7, 9\}$, and $\{4, 8, 9\}$.)

- (f) This is not a basic feasible solution, as it is not feasible.

$$\begin{pmatrix} -4 & 8 & 0 & 5 & 3 & 3 & 1 & 3 & 2 \\ 1 & -5 & 4 & 4 & 3 & 1 & 0 & 4 & 3 \\ 2 & -4 & -1 & 2 & 2 & 0 & 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 3 \\ 0 \\ 2 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 19 \\ 3 \end{pmatrix} \neq \begin{pmatrix} 9 \\ 10 \\ 6 \end{pmatrix}$$

A4-3. (a) We first convert the LP into canonical form for our basis $\{3, 5\}$.

$$\begin{aligned} \max \quad & (6 \ 5 \ 0 \ 8 \ 0) x - 5 \\ \text{s.t.} \quad & \\ & \begin{pmatrix} 2 & 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 2 & 1 \end{pmatrix} x = \begin{pmatrix} 4 \\ 5 \end{pmatrix} \\ & x \geq 0 \end{aligned}$$

The BFS is $(0, 0, 4, 0, 5)^T$.

Iteration 1. There are 3 possible candidates for entering variables: x_1, x_2, x_4 , as their coefficients in the objective function are positive. Using Bland's Rule, we pick x_1 as the entering variable. To find the leaving variable, we set $x_1 = t$, and calculate $t = \min\{\frac{4}{2}, \frac{5}{1}\} = 2$. This means x_3 is the leaving variable. The new basis is $\{1, 5\}$.

The canonical form for the new basis is

$$\begin{aligned} \max \quad & (0 \ 2 \ -3 \ 5 \ 0) x + 7 \\ \text{s.t.} \quad & \\ & \begin{pmatrix} 1 & 0.5 & 0.5 & 0.5 & 0 \\ 0 & 0.5 & -0.5 & 1.5 & 1 \end{pmatrix} x = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \\ & x \geq 0 \end{aligned}$$

The new BFS is $(2, 0, 0, 0, 3)^T$.

Iteration 2. There are 2 possible candidates for entering variables: x_2, x_4 , as their coefficients in the objective function are positive. Using Bland's Rule, we pick x_2 as the entering variable. To find the leaving variable, we set $x_2 = t$, and calculate $t = \min\{\frac{2}{0.5}, \frac{3}{0.5}\} = 4$. This means x_1 is the leaving variable. The new basis is $\{2, 5\}$.

The canonical form for the new basis is

$$\begin{aligned} \max \quad & (-4 \ 0 \ -5 \ 3 \ 0) x + 15 \\ \text{s.t.} \quad & \\ & \begin{pmatrix} 2 & 1 & 1 & 1 & 0 \\ -1 & 0 & -1 & 1 & 1 \end{pmatrix} x = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \\ & x \geq 0 \end{aligned}$$

The new BFS is $(0, 4, 0, 0, 1)^T$.

Iteration 3. There is only one candidates for entering variables: x_4 . To find the leaving variable, we set $x_4 = t$, and calculate $t = \min\{\frac{4}{1}, \frac{1}{1}\} = 1$. This means x_5 is the leaving variable. The new basis is $\{2, 4\}$.

The canonical form for the new basis is

$$\begin{aligned} \max \quad & (-1 \ 0 \ -2 \ 0 \ -3) x + 18 \\ \text{s.t.} \quad & \begin{pmatrix} 3 & 1 & 2 & 0 & -1 \\ -1 & 0 & -1 & 1 & 1 \end{pmatrix} x = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \\ & x \geq 0 \end{aligned}$$

The new BFS is $(0, 3, 0, 1, 0)^T$.

Since all the coefficients in the objective row are non-positive, we have an optimal solution $(0, 3, 0, 1, 0)^T$ with optimal value 18.

(b) We first convert the LP into canonical form for our basis $\{2, 3, 5, 6\}$.

$$\begin{aligned} \max \quad & (3 \ 0 \ 0 \ -1 \ 0 \ 0) x - 12 \\ \text{s.t.} \quad & \begin{pmatrix} -1 & 1 & 0 & -1 & 0 & 0 \\ 1 & 0 & 1 & -1 & 0 & 0 \\ -6 & 0 & 0 & 2 & 1 & 0 \\ 2 & 0 & 0 & -1 & 0 & 1 \end{pmatrix} x = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 3 \end{pmatrix} \\ & x \geq 0 \end{aligned}$$

The BFS is $(0, 2, 1, 0, 0, 3)^T$.

Iteration 1. There is only one candidates for entering variables: x_1 . To find the leaving variable, we set $x_1 = t$, and calculate $t = \min\{-, \frac{1}{1}, -, \frac{3}{2}\} = 1$. This means x_3 is the leaving variable. The new basis is $\{1, 2, 5, 6\}$.

The canonical form for the new basis is

$$\begin{aligned} \max \quad & (0 \ 0 \ -3 \ 2 \ 0 \ 0) x - 9 \\ \text{s.t.} \quad & \begin{pmatrix} 1 & 0 & 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 0 & 0 \\ 0 & 0 & 6 & -4 & 1 & 0 \\ 0 & 0 & -2 & 1 & 0 & 1 \end{pmatrix} x = \begin{pmatrix} 1 \\ 3 \\ 6 \\ 1 \end{pmatrix} \\ & x \geq 0 \end{aligned}$$

The new BFS is $(1, 3, 0, 0, 6, 1)^T$.

Iteration 2. There is only one candidates for entering variables: x_4 . To find the leaving variable, we set $x_4 = t$, and calculate $t = \min\{-, -, -, \frac{1}{1}\} = 1$. This means x_6 is the leaving variable. The new basis is $\{1, 2, 4, 5\}$.

The canonical form for the new basis is

$$\begin{aligned} \max \quad & (0 \ 0 \ 1 \ 0 \ 0 \ -2)x - 7 \\ \text{s.t.} \quad & \begin{pmatrix} 1 & 0 & -1 & 0 & 0 & 1 \\ 0 & 1 & -3 & 0 & 0 & 2 \\ 0 & 0 & -2 & 1 & 0 & 1 \\ 0 & 0 & -2 & 0 & 1 & 4 \end{pmatrix} x = \begin{pmatrix} 2 \\ 5 \\ 1 \\ 10 \end{pmatrix} \\ & x \geq 0 \end{aligned}$$

The new BFS is $(2, 5, 0, 1, 10, 0)^T$.

Iteration 3. There is only one candidates for entering variables: x_3 . However, we see that all coefficients in the column for x_3 are negative, hence the LP is unbounded.

We set $x_3 = t$, and this gives us the following set of feasible solutions:

$$x(t) = \begin{pmatrix} 2+t \\ 5+3t \\ t \\ 1+2t \\ 10+2t \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \\ 0 \\ 1 \\ 10 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 3 \\ 1 \\ 2 \\ 2 \\ 0 \end{pmatrix}.$$

So our certificate of unboundedness consists of $\bar{x} = (2, 5, 0, 1, 10, 0)^T$ and $d = (1, 3, 1, 2, 2, 0)^T$.

We see that the objective value of $x(t)$ is $t - 7$. To get a solution with objective value 9993, we want $t - 7 = 9993$, meaning $t = 10000$. So one solution is

$$(10002, 30005, 10000, 20001, 20010, 0)^T.$$